Timer Studio

Update Manual

Version 2026.05.13.0002

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Timer Studio is a multi-timer productivity app for Windows with Pomodoro mode, calendar integration, SMTP email alerts, statistics dashboard, and an always-on-top clock overlay. Supports multiple concurrent timers, recurrence, categories, hotkeys, and 5 named presets - all in a self-contained WPF application.

1 What's New in v2026.05.13.0002

This release resolves four critical bugs and six high-severity issues identified in the 2026-05-11 code audit, plus WCAG 2.2 AAA accessibility improvements.

1.1 Critical Bug Fixes

* **C-1 MediaPlayer STA thread leak** - Each custom-sound play previously leaked one STA thread and one MediaPlayer COM wrapper. Fixed: MediaPlayer.MediaEnded now calls Dispatcher.CurrentDispatcher.InvokeShutdown() then player.Close().
* **C-2 Double-dismiss race (Pomodoro / recurrence)** - _flashOverlay and _lensAlert both firing Dismissed could advance Pomodoro phase twice. Fixed: _model.AlertActive = false is now the first statement in DoDismiss() before any state mutation.
* **C-3 _calendarPollTick not stopped in OnClosed** - Poll was firing on disposed dispatcher causing ObjectDisposedException. Fixed: _calendarPollTick?.Stop() added to OnClosed.
* **C-4 TimerHistory.Record null-coalesce** - Fragile Substring on original null-possible variable corrected to: notes = notes?.Length > 200 ? notes[..200] : (notes ?? "");

1.2 High-Severity Bug Fixes

* **H-1 TimerModel._nextId atomic** - Plain int++ replaced with Interlocked.Increment; prevents duplicate IDs under concurrent timer construction.
* **H-2 HiddenCalendarPaths defensive copy** - List copied before entering Task.Run closure to prevent concurrent modification.
* **H-3 AlertSoundPlayer thread-pool starvation** - Removed outer Task.Run wrapper around thread.Join(10s); STA thread creation/join now direct.
* **H-4 SMTP password key hardened** - AES key no longer derived from UserName+MachineName. Now uses DPAPI machine-unique key.

1.3 Accessibility Improvements

* Timer card controls raised to >= 44x44px touch targets (SC 2.5.5 AAA).
* TimerCard focus ring corrected to 2px Gold/Cyan; no obscured focus states (SC 2.4.11 AAA).
* Statistics dashboard live regions wired to AutomationPeer (SC 4.1.3).
* FlashOverlay announces alert text via AutomationPeer.RaiseAutomationEvent before displaying.
* Category colour labels include accessible name + role, not colour only (SC 1.4.1).

2 System Requirements

Requirement	Minimum	Recommended
OS	Windows 10 (1809+) x64	Windows 11 x64
.NET runtime	None - self-contained	None - self-contained
RAM	64 MB	128 MB+

Requirement	Minimum	Recommended
Display	1024x768	1920x1080+
Network	None (local only)	Optional: SMTP server for email alerts

Note: No .NET runtime installation required. The executable is fully self-contained.

3 Update Instructions

3.1 Installer Update

* Close **Timer Studio** if running (check system tray).
* Download **TimerStudio_Setup.exe** from the release page.
* Run installer. Detects existing installation and upgrades in-place.
* All settings, presets, and preferences are preserved.
* On first launch a **What's New** dialog highlights changes in this version.

3.2 Portable Update

* Download **TimerStudio_Portable_<version>.zip**.
* Extract to a temporary folder.
* Copy **TimerStudio.exe** over the existing portable executable.
* Your **Data** folder (settings, presets, logs) is untouched.

3.3 Silent Update

```
TimerStudio_Setup.exe /S
```

Installs or updates silently. Suitable for managed deployment.

3.4 Rollback

Run a previous version's installer to downgrade. Settings are preserved. Versioned builds are in the **Builds** subfolder of the project.

4 Feature Overview

Feature	Detail
Multiple timers	Any number of concurrent countdown or stopwatch timers.
Pomodoro mode	Per-timer Pomodoro with configurable work, short break, and long break durations.
Recurrence	Timers auto-reset: once, N times, or indefinitely on completion.
Categories	User-defined categories with configurable colours for visual grouping.

Feature	Detail
Calendar integration	Polls Windows Calendar (CalDAV) to auto-create timers from upcoming events.
Alert styles	Flash overlay, lens popup, sound (system or custom file), or any combination.
Email alerts	SMTP email with OAuth2 Modern Auth for Outlook/Hotmail accounts.
Stats dashboard	Per-timer and aggregate stats; completion rate; active-time tracking.
Always-on-top clock	Pill overlay and full ClockWindow showing current time.
Global hotkeys	Start/stop/reset/dismiss; fully remappable via Hotkeys dialog.
5 named presets	Store full timer configurations; Ctrl+1-5 to load.
History	Session history with notes (200-char cap); accessible from Stats dashboard.
Tray icon	Minimise to tray; right-click for quick actions.

5 Accessibility (WCAG 2.2 AAA)

This application targets WCAG 2.2 Level AAA across all interactive surfaces.

Criterion	Implementation
Contrast (SC 1.4.6 AAA 7:1)	All text tokens >= 7:1 on Dark and Light themes.
Touch targets (SC 2.5.5 AAA 44px)	All timer card and toolbar controls >= 44x44px.
Focus appearance (SC 2.4.11 AAA)	2px Gold (dark) / Cyan (light) focus ring on all controls.
No colour-only info (SC 1.4.1)	Category colours paired with text labels and accessible names.
Live regions (SC 4.1.3)	Stats dashboard and alert announcements via AutomationPeer.
Keyboard navigation (SC 2.1.1)	Full Tab navigation; all dialogs keyboard-dismissible.
Focus not obscured (SC 2.4.11)	No sticky headers or overlapping panels that could hide focus ring.

6 Security Notes

SMTP credential storage was strengthened in this release (H-4 fix above). Existing stored credentials are automatically migrated to DPAPI-protected format on first launch. You will NOT need to re-enter your password unless you change user profiles or machines.

> *Note: Timer Studio makes no network calls except SMTP email alerts and optional CalDAV calendar polling. No telemetry, no analytics, no PII transmitted.*

7 Known Issues & Notes

SMTP credential bound to user profile

DPAPI key is tied to the current user profile. Re-imaging or roaming profiles require re-entering SMTP credentials.

Calendar poll ObjectDisposedException (fixed)

This was C-3. Confirmed fixed. If you experienced crashes on app exit, update to this version.

What's New dialog - shown once

To re-open: About dialog > 'What's New' button.

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This document is provided for informational purposes. Specifications subject to change.